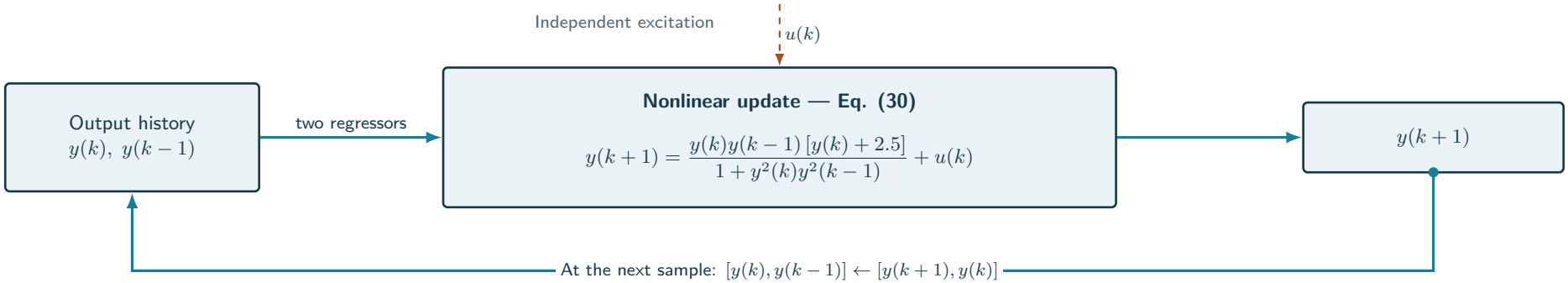
Artificial dynamic systems: two exact recurrences

Paper section 3.1: *Artificial system identification problems*

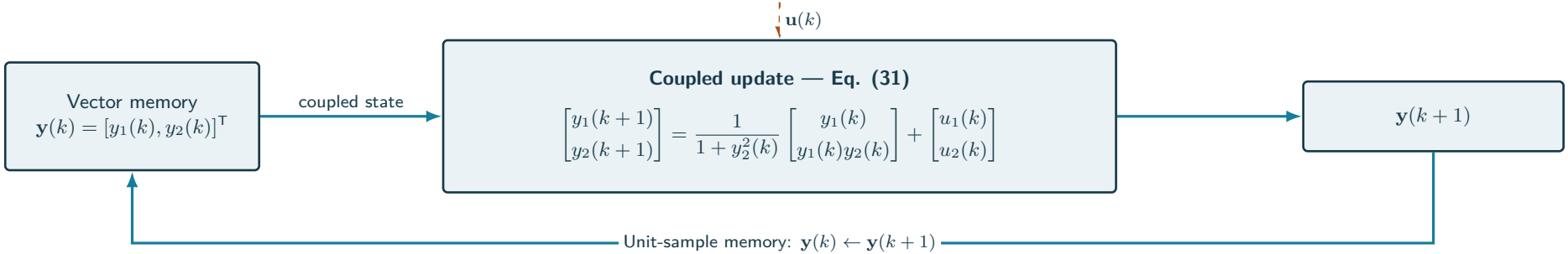
A. SISO: Narendra–Parthasarathy example 2

One input, one output; second-order recurrence



B. MIMO: Narendra–Parthasarathy example 4

Two inputs, two coupled outputs



Identification excitation

Article: 10^5 i.i.d. samples.
SISO: $u \sim \mathcal{U}[-2, 2]$.
MIMO: each $u_i \sim \mathcal{U}[-1, 1]$.

Validation excitation

SISO: $u(k) = \sin(2\pi k/25)$.
MIMO: $u_1(k) = \sin(2\pi k/25)$, $u_2(k) = \cos(2\pi k/25)$.

Evaluation contract

Distinguish measured-history one-step prediction from recursive free-run prediction. Declare initialization, sample count and seed.

Blue arrows: numerical state/history propagation. Dashed orange arrows: independent excitation. These artificial identification problems have no implied physical plant. Repository zero-history initialization and random seeds are declared experimental choices.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.